Energy mode versus two-fermion continuum: slow-mode competition in the Lindbladian of the dissipative Sachdev-Ye-Kitaev model

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The Lindbladian of the Sachdev-Ye-Kitaev (SYK) model with linear Majorana jump operators of strength μ has the Hamiltonian H as an exact eigenoperator with decay rate $q\mu$ [García-García *et al.*, Phys. Rev. E **113**, 024116 (2026)]. We ask when this energy mode sets the spectral gap of the physical, parity-even operator space. This addresses two points left open in earlier work: the relation between the finite-N gap 4μ and the large-N decay rate of fermion correlations, and the expectation that conservation laws spoil anomalous relaxation. We separate a slow sector of energy-diagonal operators from a fast sector of coherences. In the slow sector, exact diagonalization (ED) for $N \leq 24$ shows that H is the slowest mode and the next secular mode is faster by a factor that grows with N. In the fast sector, first-order perturbation theory gives rates $N\mu/2$ at finite N, whereas the large-N two-fermion threshold $2\Gamma_1$ stays finite as $\mu \to 0$. We then solve the large-N Bethe-Salpeter equation for the four-point function on the doubled contour. The singlet ladder reproduces the energy mode as a bound state at $E = q\mu$, to within 10^{-3} relative accuracy for $0.05 \leq \mu/J \leq 0.3$. In the region we resolve, we find no other singlet or non-singlet bound state below $2\Gamma_1$. For $q = 4$ this gives a gap transition at $\mu_c = 0.323J$, where $q\mu_c = 2\Gamma_1(\mu_c)$, and an analogous transition for $q = 8$. ED with explicit mode identification gives a crossing that moves from $0.086J$ ($N = 10$) to $0.184J$ ($N = 18$). We explain the slow finite-size convergence by the crossover of the fast branch from $N\mu/2$ to $2\Gamma_1$. We also show where the picture fails: it does not hold for $N \leq 8$, and it does not apply to spin chains with lower-weight conservation laws.

I. INTRODUCTION

The asymptotic relaxation rate of a Markovian open quantum system is set by the spectral gap Δ of its Lindbladian [1, 2]. For chaotic many-body systems, the weak-dissipation limit of Δ can be singular if one takes $N \to \infty$ first. This gives a finite "anomalous" relaxation rate, which has been related to quantum Ruelle-Pollicott resonances [3–6]. The SYK model [7–9] with linear jump operators $L_i = \sqrt{\mu}\,\psi_i$ is a solvable example. Its fermion Green's function decays at a rate Γ_1 that remains finite as $\mu \to 0$ [10, 11]. The Keldysh contour of this model maps onto a two-site non-Hermitian SYK model with a wormhole-like dual [11, 12].

Conserved quantities change this picture [4, 13]. A conserved operator that is also an eigenvector of the dissipator decays at a rate independent of the dynamics [14]. In the SYK model this is exact: H, the parity P and PH are simultaneous eigenvectors of the Hamiltonian and dissipative parts of the Lindbladian, with rates $q\mu$, $N\mu$ and $(N-q)\mu$ [15]. Finite-N ED at small μ correspondingly shows a gap 4μ for $q = 4$ and $N > 8$ [11]. Reference [11] left open "a more precise understanding of the relation between the spectral gap Δ and the decay rate" of the Green's function. In random open quantum circuits, Ref. [20] argued that conservation laws would spoil anomalous relaxation and deferred a systematic study. Earlier, Ref. [21] showed that a conserved magnetization makes the gap of a dissipative XXZ chain close. The exact energy mode makes the SYK model a natural place to quantify this effect.

These exact modes do not by themselves determine the gap. We address three questions. (i) Which modes compete with H, and for which N is H actually the slowest one? (ii) At large N, does any collective two-fermion mode lie below the threshold $2\Gamma_1$ set by a pair of single-fermion resonances? (iii) Where does the competition between the energy mode and the fast sector resolve, and how does the finite-N answer approach the large-N answer? Question (ii) requires the four-point ladder. The general ladder recursion on the Keldysh contour is known [28], but for the dissipative SYK model it has so far been projected only onto out-of-time-order correlators to extract Lyapunov exponents [16, 28, 29]. Here we use the time-ordered sector, whose bound states are eigenvalues of the Lindbladian.

II. MODEL, EXACT MODES, AND RIGOROUS BOUNDS

We take N Majorana fermions, $\{\psi_i, \psi_j\} = \delta_{ij}$, and $H = \sum_{i_1<\cdots<i_q} J_{i_1\ldots i_q}\psi_{i_1}\cdots\psi_{i_q}$ with $\overline{J^2_{i_1\ldots i_q}} = J^2/\binom{N-1}{q-1}$ and $q \equiv 0 \pmod 4$. This normalization gives the standard large-N coupling $\mathcal{J}^2 = qJ^2/2^{q-1}$. On parity-even operators the Heisenberg-picture Lindbladian is

$$\mathcal{L}(O) = i[H, O] - \mu\,\hat{n}(O), \qquad \hat{n}(O) = \tfrac{N}{2}O - \sum_i \psi_i O \psi_i, \tag{1}$$

where $\hat{n}$ multiplies a Majorana string by its length [17–19]. Since $\hat{n}$ is Hilbert-Schmidt Hermitian and $i[H,\cdot]$ is anti-Hermitian, every right eigenvector X obeys $-\mathrm{Re}\,\lambda_X = \mu\,(X, \hat{n}X)/(X, X)$. This is a special case of the general statement that $\mathrm{Re}\,\lambda$ lies between the extreme eigenvalues of the Hermitian part of the Liouvillian [22].

The smallest nonzero size of a traceless even operator is 2. Combined with the exact mode $\mathcal{L}(H) = -q\mu H$ [15], this gives

$$2\mu \le \Delta_N(\mu) \le q\mu \tag{2}$$

for every N and every disorder realization. The full gap therefore closes linearly in μ, whatever order of limits is taken. The exact modes are simultaneously left and right eigenvectors, so they do not hybridize with the rest of the spectrum. For instance, $\mathrm{span}\{\mathbb{1}, H\}^{\perp}$ is $\mathcal{L}$-invariant.

This does not guarantee that H is the slowest mode, and the restriction $N > 8$ noted in Ref. [11] has a simple origin. For $N = 6$ and $q = 4$, the exact mode PH [15] has rate $(N-q)\mu = 2\mu$, and Eq. (2) then forces $\Delta = 2\mu$ for all μ. We confirmed this numerically. For $N = 8$ we find that H is degenerate with the coherence sector at $\mu \to 0$, and the gap is set by coherences for all $\mu > 0$ (see below). Similarly, in spin models under uniform depolarizing noise of single-site rate γ, a two-body Hamiltonian is an eigenvector with rate 2γ. A total magnetization conserved by the Hamiltonian, as in the XXZ chain, decays at γ, so in that case the energy mode never sets the gap. This is the mechanism behind the closing gap of the dissipative XXZ chain found in Ref. [21]. The question is therefore quantitative, and we now set up the decomposition needed to answer it.

III. SLOW AND FAST SECTORS

At $\mu = 0$ the kernel of $i[H, \cdot]$ consists of energy-diagonal operators. When the spectrum within each parity block is nondegenerate, this kernel has dimension 2^{N-1}. We call it the slow sector $\mathcal{S}$; its orthogonal complement $\mathcal{F}$, the coherences, is the fast sector. Projecting out only $\mathbb{1}$ and H is not enough: all of $\mathcal{S}$ becomes slow for small μ. Mori's projection removes the full energy-diagonal subspace for the same reason [4], and the split is the weak-dissipation version of a metastable manifold [24]. To first order in μ, the rates are the eigenvalues of $\hat{n}$ restricted to each block of fixed Bohr frequency.

Slow sector. On $\mathcal{S}$, first-order theory gives the secular generator $W_{EE'} = \sum_i |\langle E|\psi_i|E'\rangle|^2 - \frac{N}{2}\delta_{EE'}$. Here E is an exact eigenvector with eigenvalue $-q$. Numerically it is the slowest nonstationary eigenvector for all $10 \le N \le 24$, and the next eigenvalue $\lambda_2 q$ is separated from it [Fig. 1(a),(b)]. In units of $q\mu$, λ_2 grows from 1.04 ($N = 10$) to 1.81 ($N = 24$). The higher eigenvalues increase with N as well. Their trend is consistent with the Ornstein-Uhlenbeck values $\lambda_k = k$, but we do not claim this limit. A controlled derivation would need concentration of $B = \sum_i [\psi_i, H]^\dagger [\psi_i, H]$, which enters through $\mathcal{L}(H^2) = -2q\mu H^2 + \mu B$. Its relative fluctuation decreases as $N^{-0.81}$ for $10 \le N \le 20$ [Fig. 1(c)], which supports the approximation but does not prove it. For the gap, the relevant fact is only that the secular modes beyond H are faster than $q\mu$ in the range we study.

Fast sector. A coherence $|E\rangle\langle E'|$ with a nondegenerate Bohr frequency has first-order rate exactly $N\mu/2$. The reason is that ψ_i is odd, so $\hat{n}$ couples such a coherence only to coherences of the opposite parity block, and generically none of those has the same frequency. Figure 1(a) confirms that for $N = 12$ and 14 the slowest fast mode has rate $N\mu/2$ to numerical precision. For $N = 10$ ($N \equiv 2 \pmod 8$) the two parity blocks have identical spectra. The resulting degenerate Bohr frequencies lower the rate to $\approx 4.55\mu$, which is still above $q\mu$. For $N = 8$ the fast rate $N\mu/2$ coincides with $q\mu$, and at small but finite μ the coherences become the slowest modes (3.5–3.8μ at $\mu = 0.02J$). The energy mode is therefore the gap at weak dissipation only for $N \ge 10$. In that regime the first-order fast rate $N\mu/2$ grows with N, whereas the large-N fast branch saturates at a finite value (next section). The fast branch must interpolate between the two. This crossover is the source of the slow finite-size convergence discussed in Sec. V.

IV. LARGE-N TWO-FERMION LADDER

Vectorizing $O \mapsto |O\rangle\!\rangle$ turns Eq. (1) into a two-site problem with fermions ψ_L and ψ_R. The dissipator becomes $\mu N/2 + i\mu \sum_i \psi_{L,i}\psi_{R,i}$, and the steady state $\mathbb{1}$ becomes the ground state [11]. Because all orders preserve the size grading, the L-R Green's functions at large N follow from the single function $C(t) = 2^{-N/2}\,\mathrm{tr}[\psi_i(t)\psi_i]$:

$$G_{LL} = G_{RR} = \mathrm{sgn}(t)C(|t|), \qquad G_{LR} = -G_{RL} = iC(|t|). \tag{3}$$

The function C obeys the Kadanoff-Baym equation

$$\dot{C} = -\mu C - 2^{2-q}J^2 \int_0^t ds\,[2C(t-s)]^{q-1}C(s), \tag{4}$$

with $C(0) = \frac{1}{2}$. We checked Eq. (4) against finite-N Heisenberg evolution: the two agree to 2×10^{-3} for $Jt \le 3$ at $N = 18$, $\mu = 0$, and at $N = 14$, $\mu = 0.2J$. The single-fermion resonance $C \sim e^{z_1 t}$ defines $\Gamma_1 = -\mathrm{Re}\, z_1$. For $q = 4$ we find $\Gamma_1(0) = 0.630J$ and $\mathrm{Im}\, z_1(0) = 0.385J$. The oscillation vanishes at an exceptional point $\mu_{\mathrm{EP}} \simeq 0.16J$, in agreement with Ref. [11].

Even operators at leading order in $1/N$ are two-fermion states. A collective mode with decay rate E appears as a normalizable solution of the Bethe-Salpeter equation $\Psi = K\Psi$. Following the Keldysh-contour recursion of Ref. [28], the ladder kernel takes the standard form [9, 16]

$$K_{ab,cd} = -(q-1)J^2 c_c c_d\, G_{ac}(t_{13})G_{bd}(t_{24})G_{cd}(t_{34})^{q-2}, \tag{5}$$

with contour signs $c_L = -i$ and $c_R = +i$. We use the ansatz $\Psi_{ab}(t_1, t_2) = e^{-E(t_1+t_2)/2}\phi_{ab}(t_1 - t_2)$. A bound state then corresponds to an eigenvalue $k(E) = 1$ of a kernel acting on functions of the relative time on $[-S, S]$. Fermionic exchange, $\phi_{ab}(s) \to -\phi_{ba}(-s)$, splits the spectrum, as in the ladder of two coupled SYK models [30]. We keep only the antisymmetric sector, which is the

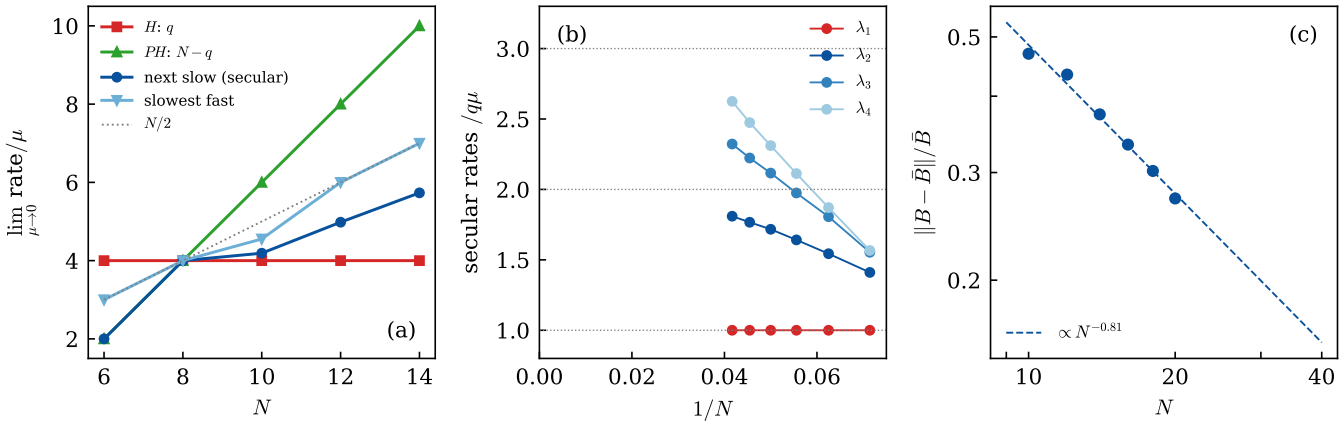


FIG. 1. Slow and fast sectors at weak dissipation ($q = 4$). (a) First-order ($\mu \to 0$) rates in units of μ: exact modes H and PH, the slowest slow-sector mode other than H, and the slowest coherence (fast) mode. Three realizations are averaged; the scatter is smaller than the symbols. Dotted: $N/2$. (b) Secular eigenvalues λ_k in units of $q\mu$; $\lambda_1 = 1$ is exact. (c) Relative Hilbert-Schmidt fluctuation of $B = \sum_i [\psi_i, H]^\dagger [\psi_i, H]$, which controls the closure of the energy-moment hierarchy.

physical one. The spurious symmetric solutions include one with $k(E) = 1$ at $E \approx 1.7\mu$ for $\mu = 0.2J$, which lies below the bound (2). This sector selection is essential: without it one would infer a nonexistent collective mode. Single-fermion poles produce a continuum starting at $2\Gamma_1$, so bound states must lie in $0 < E < 2\Gamma_1$. The non-singlet channel ($\psi_i\psi_j$, $i \neq j$) has kernel $-K/(q-1)$.

The results are shown in Fig. 2(a),(b). The leading singlet eigenvalue $k_0(E)$ increases monotonically and crosses 1 exactly once below threshold. For $0.05 \leq \mu/J \leq 0.3$ the crossing is at $E/\mu = 4.000$–4.003, converged in S for $S \geq 60$–100 and insensitive to the time step ($\delta t = 0.05$ vs 0.025). At smaller μ the deviation grows (4.016 at $\mu = 0.01J$, 4.06 at $\mu = 0.0025J$) because the bound state extends over times $\sim 1/(q\mu)$ and the relative-time grid resolves it less well. This is the exact energy mode, recovered from the large-N four-point function without using it as input. It therefore checks the kernel, the contour signs and the sector selection. The subleading eigenvalues stay below 1 throughout. A few roots of the subleading branch appear within $\lesssim 2\%$ of threshold, and they move to threshold as S grows; they are threshold artifacts. In the non-singlet channel the most negative eigenvalue of K is ≥ -0.02 for $0.05 \leq \mu/J \leq 0.5$, far from the value $-(q-1)$ needed for a bound state. A search over complex E with $\mathrm{Re}\, E < 2\Gamma_1$ found no complex root that is stable in S. Within this resolution, the only collective even mode below the two-fermion continuum is the energy mode.

For $\mu > \mu_c$, where $q\mu_c = 2\Gamma_1(\mu_c)$, the energy mode lies above threshold. The singlet root then approaches $2\Gamma_1$ as $S \to \infty$; at $\mu = 0.4J$ its distance from threshold decreases as $1/S$. The large-N gap of the even sector is thus

$$\Delta_\infty(\mu) = \min\{q\mu,\ 2\Gamma_1(\mu)\}, \qquad \mu_c = 0.323J \ (q = 4). \tag{6}$$

For $\mu > \mu_c$, Δ_∞ is the edge of a continuum rather than an isolated eigenvalue. At μ_c the derivative $d\Delta/d\mu$ jumps from 4 to $2\Gamma_1'(\mu_c) \approx 0.46$. We repeated the analysis for $q = 8$. At $\mu = 0.015J$ the bound state sits at $E/\mu = 8.000$ ($S = 300$). At $\mu = 0.025J$ and $0.03J$, which lie above the $q = 8$ crossing, the roots approach threshold as S grows ($E/\mu = 5.96, 6.18, 6.26$ for $S = 150, 300, 450$, with threshold 6.43). The crossing condition $q\mu_c = 2\Gamma_1(\mu_c)$ gives $\mu_c = 0.0199J$ for $q = 8$. At large q, where $\Gamma_1 = 2\sqrt{\mathcal{J}^2 + (q\mu/2)^2}/q$ [10], it reduces to $\mu_c = 4\mathcal{J}/(q\sqrt{q^2-4})$. The full Schwinger-Dyson value approaches this as $1 + 1.8/q$ for $q = 4$–12.

V. FINITE-N CROSSING WITH MODE IDENTIFICATION

At finite N we diagonalize $\mathcal{L}$ on $\mathrm{span}\{\mathbb{1}, H\}^\perp$ with a deflated Arnoldi method, as Eq. (1) prescribes. We classify each eigenvector by its energy-diagonal weight $w = \|\mathcal{D}(X)\|^2/\|X\|^2$, where $\mathcal{D}$ dephases in the energy eigenbasis of each parity block. We define $\mu_c(N)$ as the coupling at which the slowest mode with $w < 1/2$ reaches $q\mu$. At this point the slowest mode on $\mathrm{span}\{\mathbb{1}, H\}^\perp$ has rate $(4.00 \pm 0.02)\mu$ and $w \leq 0.15$ for every realization and every N studied. The crossing is thus between the energy mode and a coherence, and no secular mode intervenes. At smaller μ we observe intermediate weights ($0.2 < w < 0.8$). These indicate finite-N hybridization between the higher secular modes and the coherences, but they do not affect the crossing.

The crossing grows monotonically with N: $\mu_c/J = 0.086(3)$, $0.137(8)$, $0.154(2)$, $0.170(1)$ and $0.184(2)$ for $N = 10$–18 [Fig. 2(c)]. Extrapolations for $N \geq 12$ give $0.28J$ (linear in $1/N$) and $0.40J$ (linear in $1/\sqrt{N}$), which bracket the ladder value $0.323J$. The data do not fix the extrapolation form, and we do not claim quantitative finite-N confirmation. The slow drift has a natural ori-

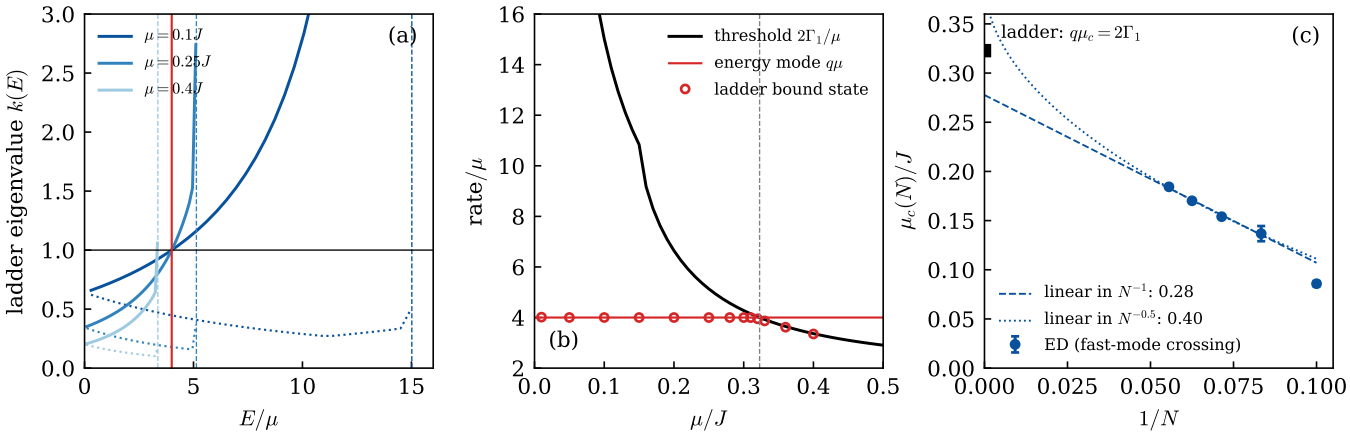


FIG. 2. (a) Leading (solid) and subleading (dotted) eigenvalues of the singlet ladder kernel in the exchange-antisymmetric sector versus trial decay rate E ($q = 4$, $S = 100$). Bound states satisfy $k(E) = 1$ (horizontal line). Dashed verticals: thresholds $2\Gamma_1$. Red vertical: $E = q\mu$. (b) Singlet bound-state rates (circles), the exact energy mode $q\mu$ (red) and the threshold $2\Gamma_1/\mu$ from Eq. (4) (black). Dashed: $\mu_c = 0.323J$. Above μ_c the circles are threshold-bound roots that converge to $2\Gamma_1$ as $S \to \infty$. (c) Finite-N crossing $\mu_c(N)$, defined as the coupling at which the slowest fast-sector mode (energy-diagonal weight $< 1/2$) reaches $q\mu$, from ED (mean $\pm$ standard error over 3–6 realizations). Lines: linear fits in $1/N$ and $1/\sqrt{N}$ for $N \geq 12$. Square: ladder value.

gin. At finite N the fast branch starts at $N\mu/2$ for $\mu \to 0$ and approaches $2\Gamma_1$ only once $N\mu$ greatly exceeds the intrinsic rate. Its value at the crossing is therefore not yet the large-N resonance. The fixed-μ rates show the same behavior: at $\mu/J = 0.4, 0.6, 0.8$, quadratic extrapolations in $1/N$ of the slowest fast rate give $1.38, 1.63, 1.93J$, compared with $2\Gamma_1 = 1.35, 1.59, 1.90J$.

VI. RELATION TO PREVIOUS GAP ANALYSES

Reference [25] computed the gap in the block of opposite parity to the steady state, i.e., among odd operators. That block contains the fermion Green's functions but not H. The authors found real "intruder" eigenvalues tied to SYK eigenstates and argued that these lose importance as $N \to \infty$, because their number scales as the Hilbert-space dimension while the full spectrum scales as its square. This is consistent with our results. A vanishing fraction of exact modes does not affect the two-point decay rate, but a single exact mode bounds the gap of the even sector, where the physical density matrix lives. Reference [25] also reported that a first-order dynamical transition of the dissipative Loschmidt echo turns into a crossover near $\mu \simeq 0.3J$, which they attribute to a proliferation of exceptional points. This value is close to our $\mu_c = 0.323J$. We have not established whether the two are related. The echo is a trace over the full operator space and is dominated by the bulk of the spectrum, whereas μ_c concerns its slowest edge. Similarly, the transitions in the dissipative form factor [26] concern bulk spectral statistics rather than the gap. The exceptional point $\mu_{\rm EP} \simeq 0.16J$ of the fermion Green's function, where the "$\mu_c \approx 0.15$" of Ref. [11] sits, is likewise distinct from μ_c.

VII. DISCUSSION

Earlier work established the exact energy mode [15] and the finite anomalous decay of fermion correlations [10, 11]. Anomalous relaxation of two-point functions also appears in an integrable, disorder-free SYK variant [27], so it is not specific to chaos. Here we determine how the energy mode competes with the rest of the even spectrum, which makes the conservation-law caveat of Ref. [20] quantitative for this model. The answer has three parts. In the slow sector, H is the slowest mode for $N \geq 10$. In the fast sector, the large-N ladder contains no collective mode below the two-fermion continuum other than H itself. The two branches cross at $q\mu_c = 2\Gamma_1(\mu_c)$. Below μ_c, relaxation is two-stage, and the gap no longer reflects the relaxation of generic observables [23]: coherences decay at a rate that remains finite as $\mu \to 0$ (at large N), while the energy relaxes at $q\mu$. The reflection of H as a bound state of the four-point ladder ties the exact Lindbladian mode to the Keldysh-wormhole description of Ref. [11]. There it corresponds to a normalizable fluctuation of the boundary energy on the doubled contour.

Our conclusions have clear limits. The absence of other bound states is numerical: it holds on the grids and box sizes we used, and it is not a proof. The finite-N crossing is consistent with the ladder value but does not pin it down. The approach of the secular rates to the Ornstein-Uhlenbeck values $\lambda_k = k$ is a trend, not a result. Beyond SYK, the mechanism requires a Hamiltonian that is an

eigenvector of the dissipator and has no conserved quantity of lower weight. As the small-N and XXZ examples show, that second condition cannot be assumed. Natural extensions are an analytic treatment of the large-q ladder, where the Liouville solution should make the bound-state condition explicit, and the double-scaled limit [31, 32].

NUMERICAL METHODS

ED uses Jordan-Wigner Majoranas and the block structure of even operators in the parity basis. Leading eigenvalues on $\mathrm{span}\{\mathbb{1}, H\}^{\perp}$ are computed with shift-deflated Arnoldi iteration (tolerance 10^{-7}). Full diagonalization was used for $N \leq 12$. First-order rates are obtained by diagonalizing $\hat{n}$ within each Bohr-frequency block (tolerance 10^{-7}). Eq. (4) is integrated with a second-order predictor-corrector scheme ($\delta t = 0.004/J$), and z_1 is obtained from the Laplace-transformed kernel. The ladder kernel is applied matrix-free by FFT convolution on $[-S, S]$ with $\delta t = 0.05/J$, using $S = 20$–$450/J$. Leading eigenvalues in the exchange-antisymmetric sector are computed by Arnoldi iteration, and roots of $k(E) = 1$ are bracketed by tracking eigenvalue branches. Code and data are available from the author.

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